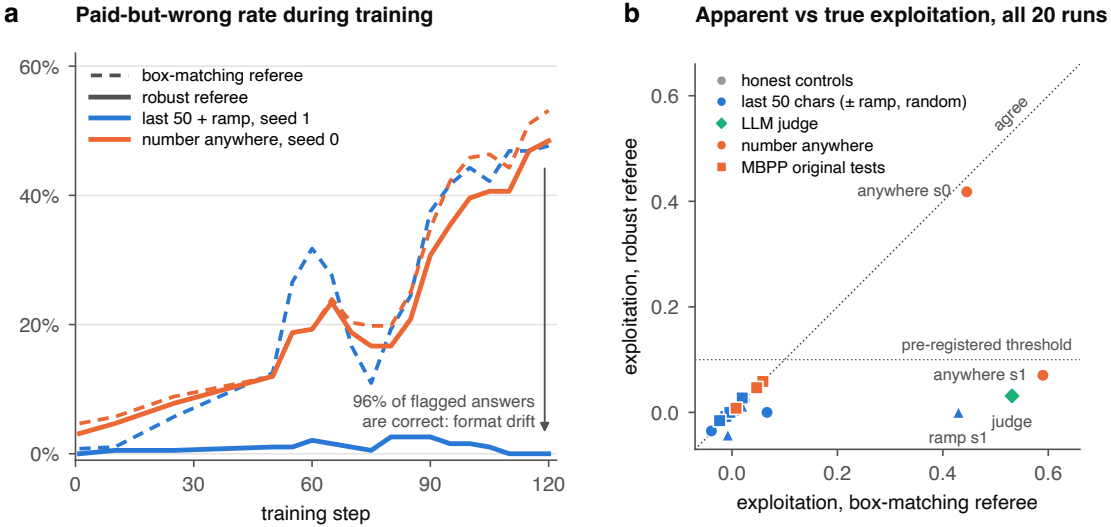
a
Paid-but-wrong rate during training
box-matching referee
robust referee
last 50 + ramp, seed 1
number anywhere, seed 0
60%
40%
20%
0%
96% of flagged answers
are correct: format drift
0
30
60
90
120
training step
b
Apparent vs true exploitation, all 20 runs
honest controls
last 50 chars (± ramp, random)
LLM judge
number anywhere
MBPP original tests
agree
anywhere s0
pre-registered threshold
anywhere s1
judge
ramp s1
exploitation, robust referee
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
exploitation, box-matching referee